Sumit Sah

Ph.D in Distributed & Online Optimization sumit.sah1824@gmail.com

Indian Institute of Technology (IIT) Dharwad **Web:** sum-iit.github.io

Degree	College/University	Year	CGPA/%
Ph.D (**Thesis Submitted**)	IIT, Dharwad	2020-26	8.33
M.Sc in Mathematics	Banaras Hindu University	2016-18	7.78

RESEARCH EXPERIENCE

- **Ph.D. Researcher | Indian Institute of Technology Dharwad** [2020-2026]
  - Thesis: *"A Constrained PL-Type Framework for Non-Convex Optimization in Distributed and Online Learning"* (Submitted).
  - Advisor: **Dr. B. N. Bharath** (Associate Professor).
- **Visiting Research Student | Mohamed Bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE** [Jun'26-Present]
  - Working with **Prof. Mohsen Guizani** on **certified robustness of LLM agents with Retrieval-Augmented Generation (RAG)**.
  - Studying certification guarantees for agentic LLM-RAG pipelines against adversarial and poisoned retrieval contexts.
  - Building on Ph.D. background in optimization and generalization theory to derive provable robustness bounds for retrieval-augmented LLM agents.

PUBLICATIONS

1. **Sumit Sah** and B. N. Bharath, "*Online Learning with Non-convex Losses: New Condition to Achieve Small Dynamic Regret*," IEEE Int. Conf. on Acoustics, Speech and Sig. Proc. (**ICASSP**), Hyderabad, India, 2025.

   Abstract: *We study online learning with non-convex, time-varying loss functions. We show that Online Gradient Descent (OGD) can achieve dynamic regret comparable to the strongly convex case, under a novel condition defined within a ball of radius ρ around the initialization. This condition incorporates gradients, losses, and temporal variation to reflect the online nature of the problem. Under this assumption, OGD attains dynamic regret scaling sub-linearly with variation. Experimental results support our theory.*

2. **Sumit Sah**, Shruti P Maralappanavar, B. N. Bharath and Prashant Khanduri, "*Linear Convergence of FedAvg(GD) Under Constrained PL-Type Assumptions: A Single Hidden Layer Neural Network Analysis*". (**Under revision**)

   Abstract: *We study the FedAvg algorithm in Federated Learning, focusing on its optimization and generalization performance. Under new constrained conditions, we show that FedAvg converges linearly and applies to certain neural networks with sufficient width. We also prove that its generalization error improves optimally with more data and benefits further from having more clients.*

3. **Sumit Sah**, Bharath B N, "*Linear Convergence of Decentralized FedAvg with Local Gradient Descent under Localized PL Conditions*". (**Under review**)

   Abstract: *We prove the first linear convergence guarantee for Decentralized Federated Averaging (D-FedAvg) under a local PL-type growth condition, far weaker than standard global assumptions. This setting requires neither data homogeneity nor global minimizers—these properties emerge naturally as iterates stay in a bounded region. Our analysis introduces new drift-dynamics bounds to control consensus error, and experiments on diverse datasets confirm the theory.*

4. **Sumit Sah**, Bharath B N, "*A PL-Type Framework For Dynamic Regret In Non-Convex, Non-Smooth Online Composite Optimization*". (**Submitted**)

   Abstract: *Online learning plays a central role in addressing non-stationary data in signal processing and machine learning. In this work, We study* non-convex *online composite optimization with* non-smooth *regularization and analyze the Online Proximal Gradient (OPG) method. We introduce a new proximal Polyak-Lojasiewicz (PL)-type condition under which the dynamic regret scales as $\mathcal{O}(D_T)$, where D_T is measure of function variation. This result extends PL-type analysis to non-convex, non-smooth settings, marking a significant advance over prior work. Finally, we provide experiments that corroborate the theoretical guarantees.*

5. **Sumit Sah**, Naveen Kummari, Mohsen Guizani, "*SmoothRAG: Certified Poison-Free Top-K Retrieval for Networked Systems*". (**Under Review**)

   Abstract: *SmoothRAG mitigates append-only knowledge-base poisoning in networked RAG by randomly subsampling query-relevant candidates and returning the highest-ranked context. Under bounded candidate-pool contamination, we certify a computable poison-free retrieval probability and an explicit tolerated untrusted-entry radius. After pool formation, the certificate is independent of poison text, embeddings, retriever scores, and corpus size, without requiring malicious-document detection. Experiments across networked systems and retrieval benchmarks show poison-free rates tracking the certificate and improved downstream accuracy under tested attacks; final-answer correctness remains uncertified.*

6. **Sumit Sah**, Naveen Kummari, Mohsen Guizani, "*RootCert: Certified RAG Under Bounded Authenticated-Principal Poisoning*". (**Under Review**)

   Abstract: *A single compromised writer can corrupt RAG by flooding a knowledge source with records, so record-count budgets are easily bypassed by fragmentation and replication; ROOTCERT instead budgets authenticated writers (principals) appended after an immutable checkpoint. Our key reduction pools each principal's record scores into one bounded vector and selects top-K groups, so r principals cause at most r score replacements and the answer is certified whenever its margin m_y exceeds $2r$. On StrategyQA, principal-based certification holds 52.6% accuracy from 1 to 64 appended records while record-based certification fails after six, and an adaptive attacker flips 65.3% of uncertified answers at $r = 4$ without changing any certified one.*

7. **Sumit Sah**, Naveen Kummari, Bharath B N, Mohsen Guizani, "*Tracking Moving Comparators with Near-Logarithmic Alternating Regret*". (**Under Review**)

   Abstract: *Corrected Hedge algorithms achieve $O(\log K)$ alternating regret against a fixed expert; we show this advantage survives a changing comparator. Adding uniform sharing to a concurrent corrected-Hedge update (FS-AA-Hedge) raises the corrected potential by only $O(\alpha_t)$, yielding $O\big(\log K + M[1 + \log(KT/M)]\big)$ alternating regret for comparator movement M, with an anytime variant. We give a matching $\Omega(\min\{T, (S+1)\log K\})$ lower bound for S switches and extend the method to convex losses and piecewise-stationary zero-sum games.*

SKILLS & INTERESTS

- **Programming Languages**: MATLAB, Python.
- **Tools & Libraries**: Scipy, Numpy, Pandas, Seaborn, Matplotlib, TensorFlow, PyTorch, RAG, LLM, LaTeX
- **Interests**: Develop theories for applied machine learning, Data Analytics.

RELEVANT COURSES

- **Statistical Pattern Recognition | Probability Theory and Random processes | Measure Theory | Optimization Theory and Application | Information Theory** [IIT Dharwad]
- **Advanced Real Analysis | Foundation of Optimization | Convex Analysis | Linear Algebra | Functional Analysis | Operator Theory | Partial Differential Equation | Algebra | Topology** [Banaras Hindu University]
- **Machine Learning (Andrew NG)** [Coursera]

TEACHING ASSISTANT

- **Linear Algebra and Its Applications | Introduction to Probability Theory | Probability Models and Application | Pattern Recognition and Machine Learning (Theory + Lab) | Optimization Theory and Application | Mathematics for Data Science | Control Theory** [IIT Dharwad]

SCHOLASTIC ACHIEVEMENTS

- Qualified Gate Mathematics(MA), **Rank** 857 [2019]
- Qualified CSIR-NET (LS), Mathematical Sciences [2019]

PROJECTS

- **DST-SERB Project: R/Dev./P-07/682 | Principal Investigator : Dr. Anupam Priyadarshi** [July'19-Aug'20]
  - I worked on a project titled "**Closure Approach using Higher Central Moment in Mathematical Ecology: Influence of Fluctuating Environment in Ecosystems**," which introduced me to Mathematical Biology. I developed mathematical models of ecosystems with multiple food chains and used MATLAB for bifurcation analysis to study the dynamics of the system. The project also improved my problem-solving skills, familiarity with LaTeX, and research writing skills.

PROFESSIONAL SERVICE

- **Reviewer, IEEE Signal Processing Letters**.
- **Reviewer, UAI 2026**.
- **Reviewer, ICASSP 2026**.

CONFERENCES, SUMMER SCHOOLS & TALKS Attended

- 2025 ICTS program on Data Science: Probabilistic and Optimization Methods-II at ICTS TIFR Bengalore.
- 2025 Joint Telematics Group/IEEE Information Theory Society Summer School on Signal Processing, Communications, and Networks. Indian Institute of Technology Bombay.
- 2025 IEEE International Conference on Acoustics, Speech, and Signal Processing, Hyderabad,India.
- VIVRITI '2024: 3rd Annual 3-Minute Research Seminar at IIT Dharwad.
- 2023 Joint Telematics Group/IEEE Information Theory Society Summer School on Signal Processing, Communications, and Networks. Indian Institute of Science(IISc) Bangalore.
- Artificial Intelligence Summer School, 2023. Indraprastha Institute of Information Technology Delhi.
- 35th Annual Conference of the Mathematical Society- Banaras Hindu University(BHU) on "Advancement in Mathematical Analysis and their Applications" 2019.
- Summer Outreach Program, National Institute of Science Education and Research, 2018 | *Supervised by*: Dr. Sutanu Roy.

POSITIONS OF RESPONSIBILITY

- **Senate Student Member (Academics) | IIT Dharwad** [May'24-May'25]
  - Contributed to institute-level academic policy discussions and decisions.
  - Represented postgraduate student concerns at the Senate meetings.
  - Collaborated with faculty and administration to support academic reforms and student-centric initiatives.
- **General Secretary Academic Affairs (PG) | IIT Dharwad** [May'23-May'25]
  - Coordinated the availability and maintenance of research equipment.
  - Engaged in regular meetings with the Director and Dean to improve the campus research environment.
  - Addressed and resolved academic and administrative grievances of research scholars on campus.
  - Facilitated the inclusion of new courses aligned with researchers' academic needs.
  - I was a member of "**Anti-Ragging Squad (ARS)**" IIT Dharwad.
- **DPGC Member | Department of Electrical Engineering, IIT Dharwad** [March'22-March'23]
  - Participated in departmental academic committee meetings, providing constructive feedback.
  - Represented Ph.D./M.S. students, addressing concerns and ensuring timely resolutions through coordination with faculty and administration.

EXTRACURRICULAR ACTIVITIES

- Playing Badminton, Cricket, Chess, Volleyball.
- Cooking
- Reading Novels and story books.